13

HOST LINK COMMUNICATION FUNCTION

This chapter describes the operating principle, communication settings, command and response of host link communication.

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13-1 Host-link Communication Function

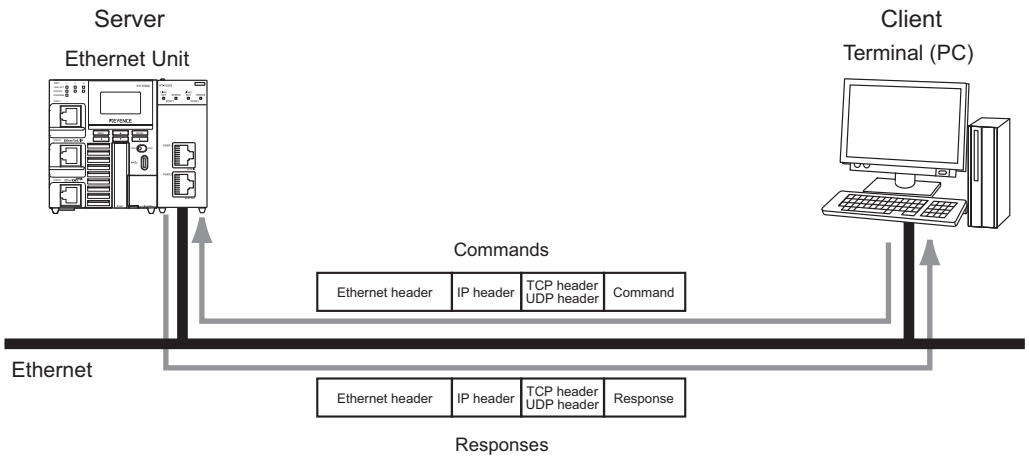
This section gives a general description on host-link communication function.

■ Application

The device values of CPU unit can be read/written by sending commands in preset format from a terminal (such as PC) connected to Ethernet.
The ladder program for communication is not required on CPU unit because a response is automatically returned to commands sent from external units.

■ Communication principle

By creating a communication program on the terminal (PC), data can be read from/written into the CPU unit, and the operation status can be monitored by sending commands to the Ethernet unit (KV-XLE02) via the Ethernet from the terminal (PC), and receiving responses via the Ethernet from the Ethernet unit.
During communication, PLC equipped with Ethernet unit serve as servers, terminals (such as PCs) serve as clients. The server automatically returns to the client the appropriate responses to commands received from the client.
Host-link communication supports TCP/IP and UDP/IP communication protocols.



13-2 Communication Specification

This section describes the communication specification of the host-link communication function. Setting of Ethernet unit terminals is executed by the Unit Editor.

Setting Items of Unit Editor

Unit Editor is used for setting up Ethernet unit communication. Setting items of Unit Editor is as follows. For the setting method, see "3-1 Unit Editor Settings", page 3-2.

Item	Settings	Setting range	Default value	See page
<Basic>				
Leading DM No.	Set No. not for any other purpose.	0 to 65234	To be set	3-8
Number of DMs in use	Number of DMs used by the unit	300 *1	300	-
Leading relay No. (set up by channel)	Set No. not for any other purpose.	0 to 1928*2	To be set	3-8
Number of relays in use	Number of relays used by the unit	1152*1	1152	-
<Port1•2>				
Setting method of IP address	Set up the IP address setting method.	It depends on the function used on the port. For details, refer to the page of each function.	Fixed IP address	3-10
IP address	Set up an IP address not duplicated with other nodes.	0.0.0.0 to 255.255.255.255	Port 1:192.168.0.10 Port 2:192.168.1.10	3-10
Subnet mask	Set up an appropriate subnet mask.	0.0.0.0 to 255.255.255.255	255.255.255.0	3-11
Communication rate	Set up according to the network used.	"1000/100/10Mbps Auto"/"100/10Mbps Auto"	1000/100/10Mbps Auto	3-11
Industrial Ethernet	Select a setting other than CC-Link IE Field (Intelligent device station) and "EtherCAT (Slave)".	Not used / EtherNet / IP (Scanner) / EtherNet / IP (Adapter) / PROFINET (Device) / CC-Link IE Field (Intelligent device station) / EtherCAT (Slave)	Not used	3-11
Host-link communication	Set up whether to use Host-link communication or not.	Enable / Disable	Enable	3-12
<Port common>				
Receive timeout[s]	Set to an appropriate value.	0～59	10	3-13
Keep-Alive [s]	Set to an appropriate value.	0 to 65535	600	3-13
Keep Alive retry interval selection	To set up Keep Alive retry interval.	Receive time out/ Keep Alive retry time	Receive time out	3-13
Keep Alive retry time [ms]	Set it to an appropriate value.	1 to 65535	10000	3-13
Default gateway	Set up an appropriate default gateway.	0.0.0.0 to 255.255.255.255	0.0.0.0	3-13
DNS server	Set up the IP address of DNS server.	0.0.0.0～ 255.255.255.255	0.0.0.0	3-15
Host-link	Change as required.	1～65535	8501	3-12

Item	Settings	Setting range	Default value	See page
<Routing setting>				
Routing setting	Set up as required.	Enable/Disable	Disable	3-17
Communication counterpart IP address 1 to 6	Set up as required.	0.0.0.0 to 255.255.255.255	0.0.0.0 (unused)	3-17
Communication counterpart subnet mask 1 to 6	Set up as required.	0.0.0.0 to 255.255.255.255	0.0.0.0 (unused)	3-17
Router IP address 1 to 6	Set up as required.	0.0.0.0 to 255.255.255.255	0.0.0.0 (unused)	3-17

*1This is the number when PROTOCOL STUDIO and KV socket communication are set to “Unused” in the unit editor.
*2 If CPU function version of KV-7000 series is 2.2 or earlier, the leading relay No. range is 0 to 928.

Communication Specification

The communication control of Ethernet unit is based on TCP/IP and UDP/IP. Specification of TCP/IP and UDP/IP communication is as follows.

■ TCP/IP

Item	Description
Communication mode	TCP/IP
Used port No.	8501 (Changeable)
Number of sockets	15(Shared with Modbus server function and MC protocol / SLMP communication function)
Data code	ASCII code

■ UDP/IP

Item	Description
Communication mode	UDP/IP
Used port No.	8501 (Changeable)
Number of sockets	1
Data code	ASCII code

Devices Used for Host-link Communication

Devices used for host-link communication function are as follows.

Item	Device No.	R/w
Send times	Leading DM+0 to 1	R
Receive times	Leading DM+2 to 3	R
Socket utilization (host-link)	Buffer memory address #9	R
Ethernet unit system error	Leading DM+20	R
MAC address (Port 1)	Leading DM+13 to 15	R
MAC address (Port 2)	Leading DM+17 to 19	R

Point

0 is stored in the error DM (leading DM + 12) if no error occurs on the Ethernet unit. In case of error, the corresponding error No. will be stored in this DM. For the setting method, see "A-1 Error List", page A-2.

13-3 Communication Procedure

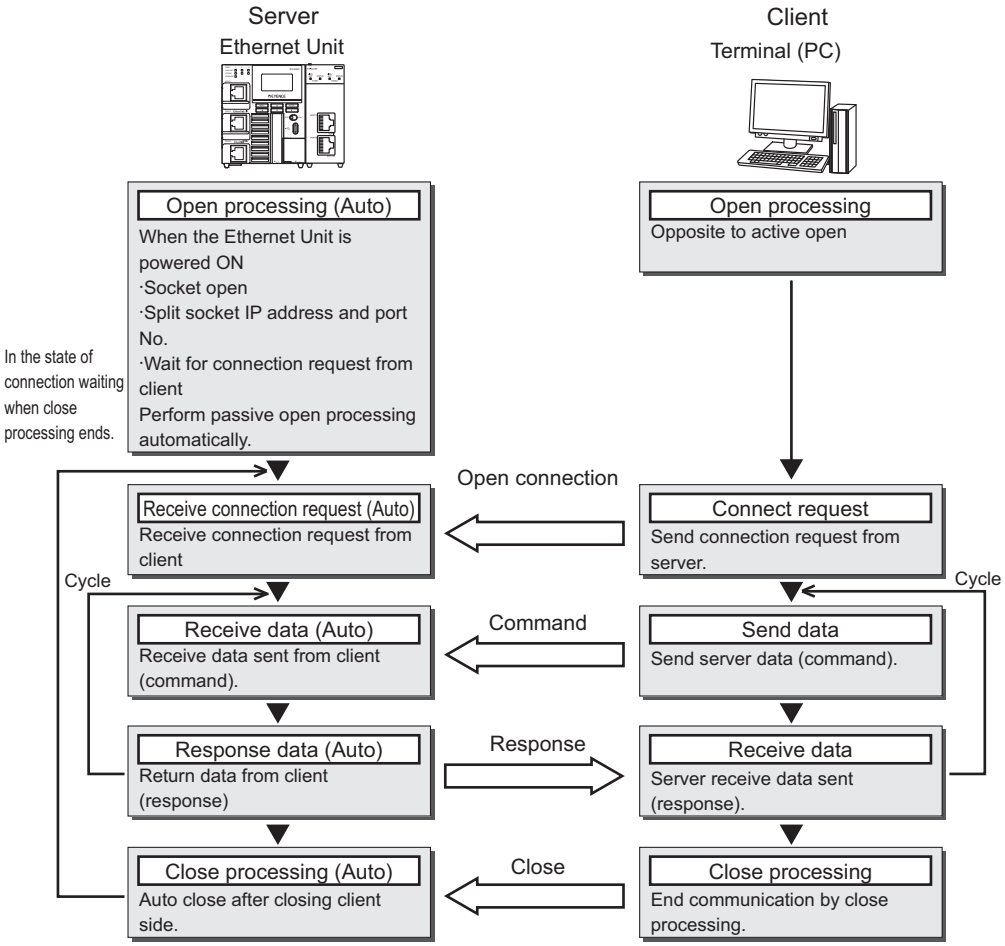
This section describes the principle of communication between Ethernet unit and a terminal (e.g. PC) on the network using the host-link communication function, as well as the format of commands and responses.

Communication Principle

Host-link communication supports TCP/IP and UDP/IP communication protocols.
The following describes the communication principles based on TCP/IP and UDP/IP protocols.

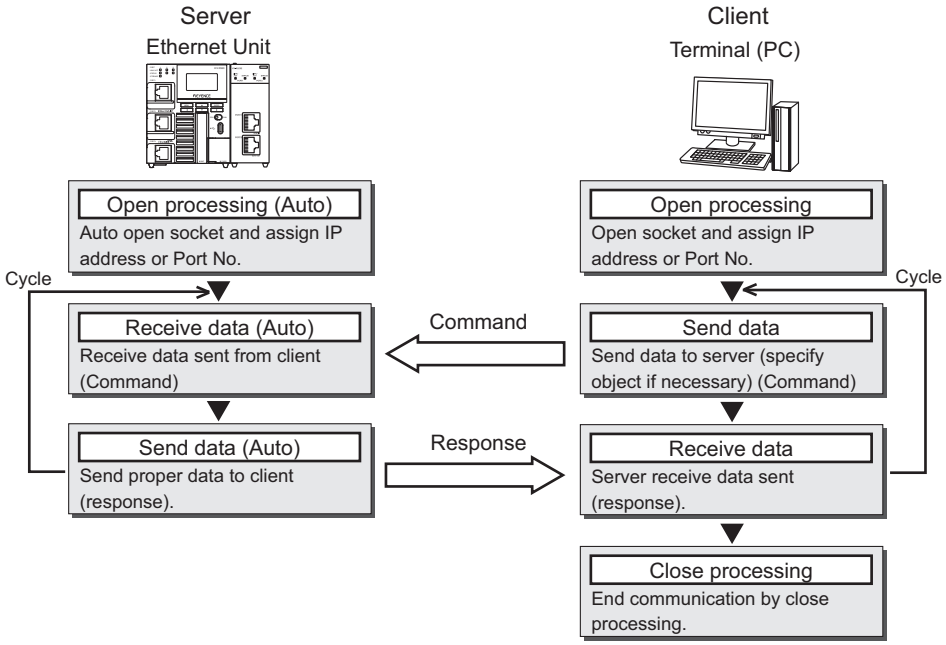
■ Communication principle based on TCP/IP protocol

Communication principle based on TCP/IP protocol is as follows.
During communication, the Ethernet unit serves as a server, and the communication destination terminal (e.g. PC) serves as a client.
The Ethernet unit automatically responds to the requests from the client.



■ Communication principle based on UDP/IP protocol

Communication principle based on UDP/IP protocol is as follows.
During communication, the Ethernet unit serves as a server, and the communication destination terminal (e.g. PC) serves as a client.
The Ethernet unit automatically responds to the requests from the client.



Command and Response Format

The following describes command and response format of communication data for host-link communication function. The format of command and response is different for TCP/IP and UDP/IP communication.

■ Format for TCP/IP communication

Command format

Send a command from the PC to the Ethernet unit according to the following format.
CR (0DH) is separator. The Ethernet unit ignores LF (0AH), characters after LF (0AH) are identified as the next command.

Ethernet header	IP header	TCP header	Command	CR

or

Ethernet header	IP header	TCP header	Command	CR	LF

Response format

The Ethernet unit returns a response to a command according to the following format. When PC generates a program, receive processing must be executed according to this response format.

Ethernet header	IP header	TCP header	Response	C_R	L_F

■ Format for UDP/IP communication

Command format

Send a command from the PC to the Ethernet unit according to the following format.
C_R (0DH) is separator. The Ethernet unit ignores L_F (0AH), characters after L_F (0AH) are identified as the next command.

Ethernet header	IP header	UDP header	Command	C_R	

or

Ethernet header	IP header	UDP header	Command	C_R	L_F

Response format

The Ethernet unit returns a response to a command according to the following format.
When PC generates a program, receive processing must be executed according to this response format.

Ethernet header	IP header	UDP header	Response	C_R	L_F

13-4 Command List

Host-link communication command list.

Function	Command	See page
Change mode	Mn	13-10
Clear error	ER	13-11
Check error No.	?E	13-11
Query model	?K	13-12
Confirm operating mode	?M	13-13
Time setting	WRT	13-14
Forced set/forced reset	ST/RS	13-15
Continuous forced set/forced reset	STS/RSS	13-16
Data read	RD	13-17
Consecutive data read	RDS	13-17
Consecutive data read	RDE	13-21
Data write	WR	13-22
Write consecutive data	WRS	13-22
Write consecutive data	WRE	13-23
Write set value*1	WS	13-24
Write consecutive set value*1	WSS	13-24
Register monitor	MBS/MWS	13-24
Read monitor	MBS/MWS	13-27
Read comments	RDC	13-28
BANK switching*1	BE	13-29
Read expansion unit buffer memory	URD	13-30
Write expansion unit buffer memory	UWR	13-31

*1 Only supported by the KV-8000/7000 series.

13-5 Description of the Commands and Responses

This section describes various commands and responses and how to set up them.

Mn Change Mode

Switches the CPU unit to PROGRAM mode or RUN mode.

■ Command

M	Mode No.	C_R
4DH		0DH

Mode No. : If "0" is specified, CPU unit is switched to PROGRAM mode.
If "1" is specified, CPU unit is switched to RUN mode.

■ Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

OK: During normal command processing, code "OK" is prompted.

■ Error response

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Instruction error

E	2	C_R	L_F
45H	32H	0DH	0AH

E2 : Program not registered

E	5	C_R	L_F
45H	35H	0DH	0AH

E5 : PLC error

" Response in case of exception", page 13-32

ER Clear Error

Removes the error in CPU unit.

■ Command

E	R	C_R
45H	52H	0DH

■ Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

OK: During normal command processing, code "OK" is prompted.

■ Error response

E	1	C_R	L_F
45H	31H	0DH	0AH

E1: Command error

" Response in case of exception", page 13-32

?E Confirm Error No.

Checks the error or exception in CPU unit. For specific error content, please check the error No. list.

■ Command

?	E	C_R
3FH	45H	0DH

■ Response

Error No.	C_R	L_F
	0DH	0AH

Error No. : Error content in CPU; error No. is prompted (no zero suppression).
If no error occurs in CPU, "000" will be displayed.

■ Error response

E	1	C_R	L_F
45H	31H	0DH	0AH

E1: Command error

" Response in case of exception", page 13-32

For error treatment method, see User's Manual of the CPU unit used.

?K Query Model

Checks PLC model.

■ Command

?	K	C_R
3FH	4BH	0DH

■ Response

Type	C_R	L_F
	0DH	0AH

Type : PLC model is displayed in "Type" in the form of No..

KV-X550	:'63'(36H, 33H)
KV-X530	:'61'(36H, 31H)
KV-X520	:'60'(36H, 30H)
KV-X500	:'62'(36H, 32H)
KV-X310	:'59'(35H, 39H)
KV-8000	:'57'(35H, 37H)
KV-8000A	:'58'(35H, 38H)
KV-7300	:'54'(35H, 34H)
KV-7500	:'55'(35H, 35H)

■ Error response

E	1	C_R	L_F
45H	31H	0DH	0AH

E1: Command error

" Response in case of exception", page 13-32

?M Confirm Operating Mode

Confirms current operating status of CPU unit.

■ Command

?	M	C_R
3FH	4DH	0DH

■ Response

Model	C_R	L_F
	0DH	0AH

Data : Displays current status (mode) of CPU unit with No. It is "0" if CPU unit is in PROGRAM mode or Ladder is not registered, and it is "1" if CPU unit is in RUN mode

■ Error response

E	1	C_R	L_F
45H	31H	0DH	0AH

E1: Command error

" Response in case of exception", page 13-32

WRT Time Setting

Sets up time of the CPU unit

■ Command

W	R	T	␣	Data 1	␣	Data 2	␣	Data 3	␣	Data 4	␣	Data 5
57H	52H	54H	20H		20H		20H		20H		20H	

␣	Data 6	␣	Data 7	C_R
20H		20H		0DH

Data 1 : Enter a two-digit value in Gregorian calendar. 00 means the year of 2000. Data range: 00 to 99.
Data 2 : Enter two-digit month. Data range: 01 to 12.
Data 3 : Enter two-digit day. Data range: 01 to 31.
Data 4 : Enter two-digit hour. Data range: 00 to 23.
Data 5 : Enter two-digit minute. Data range: 00 to 59.
Data 6 : Enter two-digit second. Data range: 00 to 59.
Data 7 : Enter one-digit week. The following table shows the relations between weekday and input value.
Data range: 0 to 6.

input value	0	1	2	3	4	5	6
week	Sun.	Mon.	Tues.	Wed.	Thu.	Fri.	Sat.

■ Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

OK: During normal command processing, code "OK" is prompted.

■ Error response

E	1	C_R	L_F
45H	31H	0DH	0AH

E1: Command error

" Response in case of exception", page 13-32

Forced Set "ST"/Forced Reset "RS"

Forcibly sets the contacts of specified device to ON/OFF.

■ Command

<Forced set>

S	T	␣	Device type	Device No.	C_R
53H	54H	20H			0DH

<Forced reset>

R	S	␣	Device type	Device No.	C_R
52H	53H	20H			0DH

Device type Device name		Device No. *1
Relay *2	R (can be omitted)	00000 to 199915 *3
Link relay	B	0000 to 7FFF
Internal auxiliary relay *2	MR	00000 to 399915
Latch relay *2	LR	00000 to 99915
Control relay	CR	0000 to 7915
Timer	T	0000 to 3999
Counter	C	0000 to 3999
Work relay*4	VB	0000 to F9FF

*1 Device No. supports zero suppression.

*2 Relays, internal auxiliary relays and latch relays can be expressed with XYM.

" XYM Expression", page 13-33

*3 If CPU function version of KV-7000 series is 2.2 or earlier, the relay range is 00000 to 99915.

*4 Can only be specified for the KV-8000/7000 series.

■ Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

OK: During normal command processing, code "OK" is prompted.

■ Error response

E	0	C_R	L_F
45H	30H	0DH	0AH

E0 : Device No. error

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Command error

" Response in case of exception", page 13-32

Continuous Forced Set "STS"/Continuous Forced Reset "RSS"

Forcibly sets the contacts of specified devices to ON/OFF.

■ Command

<Forced set>

S	T	S	␣	Device type	Device No.	␣	Number of written data	C_R
53H	54H	53H	20H			20H		0DH

<Forced reset>

R	S	S	␣	Device type	Device No.	␣	Number of written data	C_R
52H	53H	53H	20H			20H		0DH

Device type / Device name		Device No. *1	Number of written devices
Relay *2	R (can be omitted)	00000 to 199915*3	01 to 16
Link relay	B	0000 to 7FFF	01 to 16
Internal auxiliary relay *2	MR	00000 to 399915	01 to 16
Latch relay *2	LR	00000 to 99915	01 to 16
Control relay	CR	0000 to 7915	01 to 16
Work relay*4	VB	0000 to F9FF	01 to 16

*1 Device No. and number of written devices support zero suppression.

*2 Relay, internal auxiliary relay and latch relay can be expressed with XYM.

" XYM Expression", page 13-33

*3 If CPU function version of KV-7000 series is 2.2 or earlier, the relay range is 00000 to 99915.

*4 Can only be specified for the KV-8000/7000 series.

■ Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

OK: During normal command processing, code "OK" is prompted.

■ Error response

E	0	C_R	L_F
45H	30H	0DH	0AH

E0 : Device No. error

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Command error

" Response in case of exception", page 13-32

Data Read "RD"/Consecutive Data Read "RDS"

RD: Reads one data of the specified device.
RDS: Reads device data of specified quantity consecutively.

■ Command

R	D	␣	Device type	Device No.	Data format*	CR
52H	44H	20H				0DH

R	D	S	␣	Device type	Device No.	Data format*	␣	Number of read data	CR
52H	44H	53H	20H				20H		0DH

* The data format (suffix) of device is specified as .U/.S/.D/.L/.H/(not specified).
.U : Decimal, 16bit, unsigned
.S : Decimal, 16bit, signed
.D : Decimal, 32bit, unsigned
.L : Decimal, 32bit, signed
.H : Hex, 16bit
(Not specified): Depends on device type
" Data simultaneity when the data format is specified as .D/.L", page 13-33

● Devices

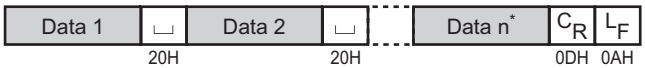
Device type		Device No. *1	Data format not specified	Number of read data	
Device name				Bit /.U/.S/.H	.D/.L
Relay*2	R (can be omitted)	00000 to 199915*3	(Bit)	0001 to 1000	0001 to 0500
Link relay	B	0000 to 7FFF	(Bit)	0001 to 1000	0001 to 0500
internal auxiliary relay*2	MR	00000 to 399915	(Bit)	0001 to 1000	0001 to 0500
Latch relay*2	LR	00000 to 99915	(Bit)	0001 to 1000	0001 to 0500
Control relay	CR	0000 to 7915	(Bit)	0001 to 1000	0001 to 0500
Work relay*6	VB	0000 to F9FF	(Bit)	0001 to 1000	0001 to 0500
Data memory*2	DM	00000 to 65534	.U	0001 to 1000	0001 to 0500
Extended data memory*2	EM	00000 to 65534	.U	0001 to 1000	0001 to 0500
File register*2	FM	00000 to 32767	.U	0001 to 1000	0001 to 0500
	ZF	00000 to 524287	.U	0001 to 1000	0001 to 0500
Link register	W	0000 to 7FFF	.U	0001 to 1000	0001 to 0500
Temporary data memory	TM	000 to 511	.U	001 to 512	001 to 256
Index register	Z	01 to 10*5	.U	01 to 12	01 to 12
Timer	T	0000 to 3999	.D	001 to 120	001 to 120
Timer (current value)	TC	0000 to 3999	.D	001 to 120	001 to 120
Timer (set value)	TS	0000 to 3999	.D	001 to 120	001 to 120
Counter	C	0000 to 3999	.D	001 to 120	001 to 120
Counter (current value)	CC	0000 to 3999	.D	001 to 120	001 to 120
Counter (set value)	CS	0000 to 3999	.D	001 to 120	001 to 120
Digital trimmer*6	AT	0 to 7	.D	1 to 8	1 to 8
Control memory	CM	0000 to 7599*4	.U	0001 to 1000	0001 to 0500
Work memory*6	VM	00000 to 589823*4	.U	0001 to 1000	0001 to 0500

*1 Device No. and number of read data support zero suppression.
*2 Relays, internal auxiliary relays, latch relays, data memories, extended data memories, and file registers can be expressed with XYM. "XYM Expression", page 13-33
*3 If CPU function version of KV-7000 series is 2.2 or earlier, the relay range is 00000 to 99915.
*4 When using the KV-7000 series, the range of CM is CM0000 to CM5999, and the range of VM is VM0000 to VM50999.

*5 For the KV-8000/7000 series, the range of Z is Z01 to Z12.
*6 Can only be specified for the KV-8000/7000 series.

■ Response

<R, B, MR, LR, CR, VB>



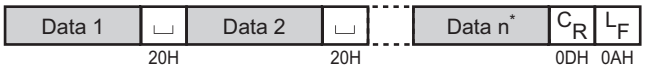
* n: Number of read data specified when sending command
Data 1 to n: The values stored in devices in turn from the specified device No..
The data range displayed varies with the data format specified when sending command.

Specified data format		Data 1 to n	
		size	range
Not specified	Bit	1byte	0 (OFF) or 1 (ON)
.U	Decimal, 16bit, unsigned	5byte	00000 to 65535
.S	Decimal, 16bit, signed	6byte	-32768 to +32767 (0 is expressed as +00000)
.D	Decimal, 32bit, unsigned	10byte	0000000000 to 4294967295
.L	Decimal, 32bit, signed	11byte	-2147483648 to +2147483647 (0 is expressed as 0000000000)
.H	Hex, 16bit	4byte	0000 to FFFF

Point

- **When specified in the data format of .U/.S/.H, the consecutive 16 bits will be processed.**
- **When specified in the data format of .D/.L, the consecutive 32 bits will be processed.**
- **If any device (R002, MR311) other than the leading device of the channel is specified for the relay device, consecutive 16/32 bits will be processed by crossing over to the next channel.**

<DM, EM, FM, ZF, W, TM, Z, AT, CM, VM>



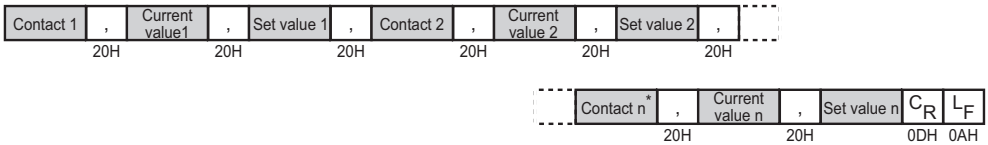
* n: Number of read data specified when sending command
Data 1 to n: The values stored in devices in turn from the specified device No..
The data range displayed varies with the data format specified when sending command.

Specified data format		Data 1 to n	
		size	range
.U (can be omitted)	Decimal, 16bit, unsigned	5byte	00000 to 65535
.S	Decimal, 16bit, signed	6byte	-32768 to +32767 (0 is expressed as +00000)
.D	Decimal, 32bit, unsigned	10byte	0000000000 to 4294967295
.L	Decimal, 32bit, signed	11byte	-2147483648 to +2147483647 (0is expressed as 0000000000)
.H	Hex, 16bit	4byte	0000 to FFFF

Point

- **If Z is specified in the format of .U/.S/.H, the value of lower 16 bits will be displayed.**
- **If AT is specified in the format of .U/.S/.H, the value of lower 16 bits will be displayed.**

<T, C>



* n: Number of read data specified when sending command
Contact 1 to n, current value 1 to n, set value 1 to n
: Displays the values stored in device in turn from the specified device No..
The data range displayed varies with the data format specified when sending command.
(For CTH, values of contact and set value are stored as fixed value 0)

Specified data format		Data 1 to n		Current value 1 to n, set value 1 to n	
		size	range	size	range
.U	Decimal, 16bit, unsigned	1byte	0 (OFF) or 1 (ON)	5byte	00000 to 65535
.S	Decimal, 16bit, signed			6byte	-32768 to +32767 (0 is expressed as +00000)
.D (can be omitted)	Decimal, 32bit, unsigned			10byte	0000000000 to 4294967295
.L	Decimal, 32bit, signed			11byte	-2147483648 to +2147483647 (0 is expressed as 0000000000)
.H	Hex, 16bit			4byte	0000 to FFFF

Point

When specified in the data format of .U/.S/.H, the value of lower 16 bits will be displayed.

<TC, CC>

* n: Number of read data specified when sending command
Data 1 to n: the values stored in devices in turn from the specified device No..
The data range displayed varies with the data format specified when sending command.

Specified data format		Data 1 to n	
		size	range
.U	Decimal, 16bit, unsigned	5byte	00000 to 65535
.S	Decimal, 16bit, signed	6byte	-32768 to +32767 (0 is expressed as +00000)
.D (can be omitted)	Decimal, 32bit, unsigned	10byte	0000000000 to 4294967295
.L	Decimal, 32bit, signed	11byte	-2147483648 to +2147483647 (0 is expressed as 0000000000)
.H	Hex, 16bit	4byte	0000 to FFFF

Point

If the format of .U/.S/.H is specified, the value of lower 16 bits will be displayed.

<TS, CS>

Set value 1	␣	Set value 2	␣		Set value n*	C_R	L_F
	20H		20H			0DH	0AH

* n: Number of read data specified when sending command

Set value 1 to n: Displays the values stored in device in turn from the specified device No.. The data range displayed varies with the data format specified when sending command.

Specified data format		Set value 1 to n	
		size	range
.U	Decimal, 16bit, unsigned	5byte	00000 to 65535
.S	Decimal, 16bit, signed	6byte	-32768 to +32767 (0 is expressed as +00000)
.D (can be omitted)	Decimal, 32bit, unsigned	10byte	0000000000 to 4294967295
.L	Decimal, 32bit, signed	11byte	-2147483648 to +2147483647 (0 is expressed as 0000000000)
.H	Hex, 16bit	4byte	0000 to FFFF

Point

When specified in the data format of .U/.S/.H, the value of lower 16 bits will be displayed.

■ Error response

E	0	C_R	L_F
45H	30H	0DH	0AH

E0 : Device No. error

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Command error

" Response in case of exception", page 13-32

■ Example of commands and responses

● Read R100 to R103 in the data format of "bit"

Command

R	D	S	␣	R	1	0	0	␣	4	C_R
52H	44H	53H	20H	52H	31H	30H	30H	20H	34H	0DH

Response

1	␣	0	␣	1	␣	0	C_R	L_F
31H	20H	30H	20H	31H	20H	30H	0DH	0AH

R100:ON R101:OFF R102:ON R103:OFF

● Read DM200 to DM202 in the data format of .S

Command

R	D	S	␣	D	M	2	0	0	.	S	␣	3	C_R
52H	44H	53H	20H	44H	4DH	32H	30H	30H	2EH	53H	20H	33H	0DH

Response

+	1	5	0	2	5	␣	−	2	5	4	0	0	␣	+	0	0	0	0	0	C_R	L_F
2BH	31H	35H	30H	32H	35H	20H	2DH	32H	35H	34H	30H	30H	20H	2BH	30H	30H	30H	30H	30H	0DH	0AH

DM200 = +15025 DM201 = -25400 DM202 = 0

Read Consecutive Data "RDE"

RDE command is compatible with Ethernet unit KV-LE20A. Its operation is the same as RDS.

"Data Read "RD"/Consecutive Data Read "RDS"", page 13-17

Write Data " WR"/Write Consecutive Data "WRS"

WR : Writes data to specified device.
WRS : Writes specified number of consecutive data.

■ Command

W	R	␣	Device type	Device No.	Data format*	␣	Data	CR
57H	52H	20H				20H		0DH

W	R	S	␣	Device type	Device No.	Data format*	␣	Number of written data	␣	Data 1	␣	Data 2		Data n	CR
57H	52H	53H	20H				20H		20H		20H				0DH

* The data format (suffix) of device is specified as .U/.S/.D/.L/.H/(not specified).
.U : Decimal, 16bit, unsigned
.S : Decimal, 16bit, signed
.D : Decimal, 32bit, unsigned
.L : Decimal, 32bit, signed
.H : Hex, 16bit
(Not specified): Depends on device type
" Data simultaneity when the data format is specified as .D/.L", page 13-33

Reference For individual device specifying method for "WR"/"WRS"command, see page 13-18.

● About devices

Device type		Device No. *1	Data format not specified	Number of written devices	
Device name				Bit /.U/.S/.H	.D/.L
Relay *2	R (can be omitted)	00000 to 199915*5	(Bit)	0001 to 1000	0001 to 0500
Link relay	B	0000 to 7FFF	(Bit)	0001 to 1000	0001 to 0500
Internal auxiliary relay*2	MR	00000 to 399915	(Bit)	0001 to 1000	0001 to 0500
Latch relay *2	LR	00000 to 99915	(Bit)	0001 to 1000	0001 to 0500
Control relay	CR	0000 to 7915	(Bit)	0001 to 1000	0001 to 0500
Work relay*8	VB	0000 to F9FF	(Bit)	0001 to 1000	0001 to 0500
Data memory *2	DM	00000 to 65534	.U	0001 to 1000	0001 to 0500
Extended data memory *2	EM	00000 to 65534	.U	0001 to 1000	0001 to 0500
File register *2	FM	00000 to 32767	.U	0001 to 1000	0001 to 0500
	ZF	00000 to 524287	.U	0001 to 1000	0001 to 0500
Link register	W	0000 to 7FFF	.U	0001 to 1000	0001 to 0500
Temporary data memory	TM	000 to 511	.U	001 to 512	001 to 256
Index register	Z	01 to 10*7	.U	01 to 12	01 to 12
Timer (current value)*4	T*3, TC	0000 to 3999	.D	001 to 120	001 to 120
Timer (set value)*4	TS	0000 to 3999	.D	001 to 120	001 to 120
Counter (current value)*4	C*3, CC	0000 to 3999	.D	001 to 120	001 to 120
Counter (set value)*4*8	CS	0000 to 3999	.D	001 to 120	001 to 120
Control memory	CM	0000 to 7599*6	.U	0001 to 1000	0001 to 0500
Work memory*8	VM	00000 to 589823*6	.U	0001 to 1000	0001 to 0500

*1 Device No. and number of written devices support zero suppression.
*2 Relays, internal auxiliary relays, latch relays, data memories, extended data memories, and file registers can be expressed with XYM. "XYM Expression", page 13-33
*3 If timers and counters are specified, the current value will be written.
*4 If the data format of .U/.S/.H is specified, data will be written in lower 16 bits.
*5 If CPU function version of KV-7000 series is 2.2 or earlier, the relay range is 00000 to 99915.
*6 When using the KV-7000 series, the range of CM is CM0000 to CM5999, and the range of VM is VM0000 to VM63999.

*7 For the KV-8000/7000 series, the range of Z is Z01 to Z12.
*8 Can only be specified for the KV-8000/7000 series.

■ Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

OK: During normal command processing, code "OK" is prompted.

■ Error response

E	0	C_R	L_F
45H	30H	0DH	0AH

E0 : Device No. Error

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Command Error

E	4	C_R	L_F
45H	34H	0DH	0AH

E4 : Write Protected

" Response in case of exception", page 13-32

■ Example of commands and responses

● Write in R100 to R103 in the data format of "bit"

Command

W	R	S	␣	R	1	0	0	␣	4	␣	1	␣	0	␣	1	␣	0	C_R
57H	52H	53H	20H	52H	31H	30H	30H	20H	34H	20H	31H	20H	30H	20H	31H	20H	30H	0DH

Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

● Write in DM200 to DM202 in the data format of .S

Command

W	R	S	␣	D	M	2	0	0	.	S	␣	3	␣
57H	52H	53H	20H	44H	4DH	32H	30H	30H	2EH	53H	20H	33H	20H

+	1	5	0	2	5	␣	−	0	5	4	0	0	␣	2	0	0	C_R
2BH	31H	35H	30H	32H	35H	20H	2DH	30H	35H	34H	30H	30H	20H	32H	30H	30H	0DH

Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

Reference Written data support zero suppression. For positive data, "+" can be omitted.

Write Consecutive Data "WRE"

WRE command is compatible with Ethernet unit KV-LE20A. Its operation is the same as WRS.

"Write Data " WR"/Write Consecutive Data "WRS"", page 13-22

Write Set Value "WS"/Write Consecutive Set Value "WSS"

WS/WSS command is compatible with Ethernet unit KV-LE20A. Its operation is the same as WR/WRS. However, "Device type" can only be specified as "T" or "C". When "T" or "C" is specified, the set value will be written.

"Write Data " WR"/Write Consecutive Data "WRS"", page 13-22

Point

When connected to a CPU unit other than the KV-8000/7000 series, the setting value writing command [WS] and the continuous setting value writing command [WSS] are not supported.

When a setting value writing [WS] command or continuous setting value writing [WSS] command is executed, an abnormal response [E1] is returned.

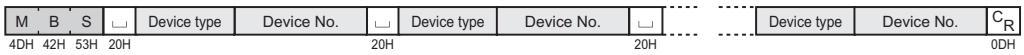
Register Monitor "MBS"/"MWS"

Registers the specified device to bit device register table (MBS) or word device register table (MWS). At most 120 data can be registered.

MBS : Register bit device in monitor.
MBS : Register word device in monitor.

■ Command

<Register bit device in monitor>



Device type Device name		Device No. *1
Relay *2	R (can be omitted)	00000 to 199915*3
Link relay	B	0000 to 7FFF
Internal auxiliary relay*2	MR	00000 to 399915
Latch relay *2	LR	00000 to 99915
Control relay	CR	0000 to 7915
Timer	T	0000 to 3999
Counter	C	0000 to 3999
Work relay*4	VB	0000 to F9FF

*1 Device No. supports zero suppression.

*2 Relays, internal auxiliary relays, latch relays, data memories, extended data memories, and file registers can be expressed with XYM.

" XYM Expression", page 13-33

*3 If CPU function version of KV-7000 series is 2.2 or earlier, the relay range is 00000 to 99915.

*4 Can only be specified for the KV-8000/7000 series.

<Register bite device in monitor>

M	W	S	␣	Device type	Device No.	Data format*	␣	Device type	Device No.	Data format*	␣	...
4DH	57H	53H	20H				20H				20H	

...	Device type	Device No.	Data format*	C_R
				0DH

* The data format (suffix) of device is specified as .U/.S/.D/.L/.H/(not specified).

.U : Decimal, 16bit, unsigned
.S : Decimal, 16bit, signed
.D : Decimal, 32bit, unsigned
.L : Decimal, 32bit, signed
.H : Hex, 16bit

(Not specified): depends on device type

" Data simultaneity when the data format is specified as .D/.L", page 13-33

● About devices

Device type / Device name		Device No. *1	Data format not specified
Relay *2*3	R (can be omitted)	00000 to 199915*4	.U
Link relay *3	B	0000 to 7FFF	.U
Internal auxiliary relay *2*3	MR	00000 to 399915*4	.U
Latch relay *2*3	LR	00000 to 99915	.U
Control relay *3	CR	0000 to 7915	.U
Work relay *6	VB	0000 to F9FF	.U
Data memory *2	DM	00000 to 65534	.U
Extended data memory *2	EM	00000 to 65534	.U
File register *2	FM	00000 to 32767	.U
Link register	W	0000 to 7FFF	.U
Temporary data memory	TM	000 to 511	.U
Index register	Z	01 to 10 *7	.U
Timer (current value)	TC	0000 to 3999	.D
Timer (set value)	TS	0000 to 3999	.D
Counter (current value)	CC	0000 to 3999	.D
Counter (set value)	CS	0000 to 3999	.D
Control memory	CM	0000 to 7599*5	.U
Work memory *6	VM	0000 to 589823*5	.U

*1 Device No. supports zero suppression.

*2 Relays, internal auxiliary relays, latch relays, data memories, extended data memories, and file registers can be expressed with XYM.

"XYM Expression", page 13-33

*3 Relays, link relays, internal auxiliary relays, latch relays, and control relays are registered by channel. Leading device should be specified when setting.

*4 If CPU function version of KV-7000 series is 2.2 or earlier, the relay range is 00000 to 99915.

*5 When using the KV-7000 series, the range of CM is CM0000 to CM5999, and the range of VM is VM0000 to VM50999.

*6 Can only be specified for the KV-8000/7000 series.

*7 For the KV-8000/7000 series, the range of Z is Z01 to Z12.

■ Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

OK: During normal command processing, code "OK" is prompted.

■ Error response

E	0	C_R	L_F
45H	30H	0DH	0AH

E0 : Device No. error

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Command error

" Response in case of exception", page 13-32

Read Monitor "MBR"/"MWR"

Reads the device value in device register table.
MBR : Reads the value of bit device registered in monitor.
MWR : Reads the value of word device registered in monitor.

■ Command

<To read bit device register table. >

M	B	R	C_R
4DH	42H	52H	0DH

<To read word device register table. >

M	W	R	C_R
4DH	57H	52H	0DH

■ Response

Data	␣	Data	␣	···	␣	Data	␣	Data	C_R	L_F
	20H		20H		20H		20H		0DH	0AH

Data : Enters the monitor result of the device registered in bit device register table or word device register table. The format of monitor result is the same as that of response to RDS command.

■ Error response

E	1	C_R	L_F
45H	31H	0DH	0AH

E1: Command error

" Response in case of exception", page 13-32

Read Comments "RDC"

Reads comments of specified device.

■ Command

R	D	C	␣	Device type	Device No.	C_R
52H	44H	43H	20H			0DH

Device type Device name		Device No. *1
Relay *2	R (can be omitted)	00000 to 199915*3
Link relay	B	0000 to 7FFF
Internal auxiliary relay*2	MR	00000 to 399915
Latch relay *2	LR	00000 to 99915
Control relay	CR	0000 to 7915
Data memory *2	DM	00000 to 65534
Extended data memory *2	EM	00000 to 65534
File register *2	FM	00000 to 32767
	ZF	000000 to 524287
Link register	W	0000 to 7FFF
Temporary data memory	TM	000 to 511
Index register	Z	01 to 10*5
Timer	T	0000 to 3999
Counter	C	0000 to 3999
Control memory	CM	0000 to 7599*4

*1 Device No. supports zero suppression.
*2 Relays, internal auxiliary relays, latch relays, data memories, extended data memories, and file registers can be expressed with XYM.
" XYM Expression", page 13-33
*3 If CPU function version of KV-7000 series is 2.2 or earlier, the relay range is 00000 to 99915.
*4 When using the KV-7000 series, the range of CM is CM0000 to CM5999.
*5 For the KV-8000/7000 series, the range of Z is Z01 to Z12.

■ Response

Data	C_R	L_F
	0DH	0AH

Data : Enters the device comments specified by command (32 characters). If the comments are less than 32 characters, space (20H) should be added. If the comment size is 32 characters or more, up to 32 characters can be entered.
The converted strings will be added according to the project language setting.

■ Error response

E	0	C_R	L_F
45H	30H	0DH	0AH

E0 : Device No. error

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Command error

E	6	C_R	L_F
45H	36H	0DH	0AH

E6 : No comment

" Response in case of exception", page 13-32

Point

In case of writing under RUN, a response to exception “E6” will be added even if comments are registered in the specified device,

BANK Switching "BE"

Switches the bank of file register.

■ Command

B	E	␣	Bank No.	C_R
42H	45H	20H		0DH

BANK No.: BANK No. of file register should be specified within the range of 0 to 15.

■ Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

OK: During normal command processing, code "OK" is prompted.

■ Error response

E	0	C_R	L_F
45H	30H	0DH	0AH

E0 : Device No. error

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Command error

" Response in case of exception", page 13-32

Point

When connected to a CPU unit other than the KV-8000/7000 series, the bank switch [BE] command is not supported.
When the bank switch [BE] command is executed, an abnormal response [E1] is returned.

Read Expansion Unit Buffer Memory "URD"

Reads expansion unit buffer memory data of specified quantity consecutively.

■ Command

U	R	D	␣	Unit No.*1	␣	Address*2	Data format*3	␣	Number of read data*4	C_R
55H	52H	44H	20H		20H			20H		0DH

*1 Unit No. : Unit No. should be specified within the range of 00 to 48.

*2 Address : The address of expansion unit buffer memory should be specified within the range of 0 to 59999.

*3 Data format : The data format (suffix) of device should be specified as .U/.S/.D/.L/.H/(not specified).

- .U : Decimal, 16bit, unsigned
- .S : Decimal, 16bit, signed
- .D : Decimal, 32bit, unsigned
- .L : Decimal, 32bit, signed
- .H : Hex, 16bit

(Not specified): Depends on device type

" Data simultaneity when the data format is specified as .D/.L", page 13-33

The method to specify data format is the same as RD/RDS command.

"Data Read "RD"/Consecutive Data Read "RDS"", page 13-17

*4 Number of read data : If the data format is specified as .U/.S/.H, it should be specified within the range of 0001 to 1000.
If the data format is specified as .D/.L, it should be specified within the range of 0001 to 0500.

■ Response

Data	␣	Data	␣	...	␣	Data	␣	Data	C_R	L_F
	20H		20H		20H		20H		0DH	0AH

Data : The data stored in the expansion unit buffer memory of specified address. The format of read result is the same as that of response of RDS command.

■ Error response

E	0	C_R	L_F
45H	30H	0DH	0AH

E0 : Device No. error

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Command error

" Response in case of exception", page 13-32

Write Expansion Unit Buffer Memory "UWR"

Writes specified number of consecutive data into the expansion unit buffer memory.

■ Command

U	W	R	␣	Unit No.*1	␣	Address*2	Data format*3	␣	Number of read data*4	␣	Data 1*5	␣	Data 2*5	␣	...	Data n*5	C_R
55H	57H	52H	20H		20H			20H		20H		20H		20H			0DH

*1 Unit No. : Specifies the unit No. within the range of 00 to 48.

*2 Address : The address of expansion unit buffer memory should be specified within the range of 0 to 59999.

*3 Data format : The data format (suffix) of device should be specified as .U/.S/.D/.L/.H/(not specifies).

- .U : Decimal, 16bit, unsigned
- .S : Decimal, 16bit, signed
- .D : Decimal, 32bit, unsigned
- .L : Decimal, 32bit, signed
- .H : Hex, 16bit

(Not specified): Depends on device type

" Data simultaneity when the data format is specified as .D/.L", page 13-33

The method to specify data format is the same as WR/WRS command.

"Write Data " WR"/Write Consecutive Data "WRS"", page 13-22

*4 Number of written data: If the data format is specified as .U/.S/.H, it should be specified within the range of 0001 to 1000.
If the data format is specified as .D/.L, it should be specified within the range of 0001 to 0500.

*5 Data 1 to n : Specifies the data to be written in the expansion unit buffer memory.
(n represents number of written data)

■ Response

O	K	C_R	L_F
4FH	4BH	0DH	0AH

OK: During normal command processing, code "OK" is prompted.

■ Error response

E	0	C_R	L_F
45H	30H	0DH	0AH

E0 : Device No. error

E	1	C_R	L_F
45H	31H	0DH	0AH

E1 : Command error

" Response in case of exception", page 13-32

Response in case of exception

A response different from normal one will be returned if a wrong command is sent or an exception occurs in the CPU unit. Response details, causes and measures to be taken in case of exception are shown below.

Code	Description	Cause	Measures
E0	Abnormal device No.	• The specified device No., BANK No., unit No. or address is out of range. • The specified timer, counter, CTH or CYC No. is unavailable in the program. • Read monitor is executed without registering in monitor.	• Specify a No. within the given range. • Check program and specify a No. used in the program. • Register in monitor before executing read monitor.
E1	Abnormal command	• Unsupported command is sent. • The method to specify command is incorrect.	• Send correct command after checking the command.
E2	Program not registered	• M1 command (switch to RUN mode) is sent in the condition that no program is registered in CPU unit. • M1 command (switch to RUN mode) is sent in the condition that the RUN/PROG switch of CPU unit is in PROG mode.	• Send the command after program is registered in CPU unit. • Switch the RUN/PROG switch of CPU unit to RUN mode.
E4	Write disabled	• You cannot change set values of timer, counter against program to which write is disabled.	• Send the command again after program write is enabled.
E5	Unit error	• "Command M1 (switch to RUN mode)" is sent when the error of CPU unit is not removed.	• Send the command again after the CPU unit error is removed.
E6	No comments	• Comments are not registered in the selected device of read comments command (RDC). • Failed to read the comments due to high load of PLC.	• If required, register the comments in device. • Execute when the load of PLC is lower or try again.

XYM Expression

For relay, internal auxiliary relay, latch relay, data memory, extended data memory and file register, devices can be specified based on XYM expression in the following table.

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Device name	KEYENCE expression	XYM expression	Device No. for XYM expression
Relay	R	X	0000 to 1999F *
		Y	0000 to 1999F *
Internal auxiliary relay	MR	M	00000 to 63999
Latch relay	LR	L	00000 to 15999
Data memory	DM	D	00000 to 65534
Extended data memory	EM	E	00000 to 65534
File register	FM	F	00000 to 32767

* If CPU function version of KV-7000 series is 2.2 or earlier, the relay range is 0000 to 999F.

Data simultaneity when the data format is specified as .D/.L

If the data format of R, B, MR, LR, CR, DM, EM, FM, ZF, W, TM, CM, VB, and VM is specified as .D/.L, the device of specified device No. is lower 16 bits, the device of next No. is upper 16 bits, and the device will be processed as 32-bit data.
Only when the device No. is an even number, the simultaneity of upper 16 bits and lower 16 bits can be ensured.